HomeWork-4 Report

Introduction

The objective of this assignment is to understand how the search algorithms like PageRank and HITS work. These algorithms are important and relevant to search engines as they can provide another level of metrics to see if the documents are relevant to the queries. Additionally, having these ranks can help reduce documents to work with; we can search for the documents that have higher importance so that we don't have to rank the entirety of our document set.

Methodology

There are two data sets: WT2G_inlinks and the merged data that our team has collected. For the WT2G_inlinks dataset, I had to create an outline list based on the inlinks. To process the merged data, I went through each document and made sure the outlinks and inlinks were present in the documents. Also, I saved the stored data with *pickle* so that it is easily accessible. To analyze the data, I used PageRank and HITS. For PageRank, I used iterative methods provided by the pseudocode in the assignment. Then for HITS, I got documents based on the queries, and I updated the authority and hub according to all pages that the page points to and a set of pages that point to the page. Also, I had to normalize the summation.

Analysis

PageRank Analysis

PageRank on WT2G_Inlinks Data

- Include a screenshot of the first 20 rows.
- Discuss any notable patterns or anomalies observed in the PageRank scores compared to inlink counts.

```
1. WT21-B37-76: inlinks - 2568, outlinks - 5, score: 0.002694470870550098
2. WT21-B37-75: inlinks - 1704, outlinks - 1, score: 0.0015331771248556479
3. WT25-B39-116: inlinks - 169, outlinks - 1, score: 0.0014685088124133142
4. WT23-B21-53: inlinks - 198, outlinks - 1, score: 0.0013735336119119804
5. WT24-B26-10: inlinks - 291, outlinks - 1, score: 0.001276215242847925
6. WT24-B40-171: inlinks - 270, outlinks - 209, score: 0.0012452589803263559
7. WT23-B39-340: inlinks - 274, outlinks - 395, score: 0.0012428610383037005
8. WT23-B37-134: inlinks - 207, outlinks - 2, score: 0.001205427413207439
9. WT08-B18-400: inlinks - 990, outlinks - 0, score: 0.0011447764346449339
10. WT13-B06-284: inlinks - 454, outlinks - 2, score: 0.0011365503403035963
11. WT13-B06-273: inlinks - 452, outlinks - 11, score: 0.0010549175456408312
12. WT01-B18-225: inlinks - 1137, outlinks - 0, score: 0.0009553812179191713
13. WT04-B27-720: inlinks - 291, outlinks - 27, score: 0.0009409558792873439
14. WT24-B26-46: inlinks - 179, outlinks - 3, score: 0.0008622309394693758
15. WT23-B19-156: inlinks - 364, outlinks - 12, score: 0.0008250669167279344
16. WT04-B30-12: inlinks - 241, outlinks - 8, score: 0.0008166209010142491
17. WT25-B15-307: inlinks - 605, outlinks - 8, score: 0.0007972229995177289
18. WT07-B18-256: inlinks - 169, outlinks - 169, score: 0.0007750022278949 52
19. WT24-B40-167: inlinks - 153, outlinks - 152, score: 0.00070760554822367
20. WT14-B03-220: inlinks - 163, outlinks - 162, score: 0.0006988599395797161
```

From the result, we can see that links with higher inlinks tend to have higher PageRank. However, there are some cases where the number of inlinks is lower but has a higher PageRank. This is because the incoming links of that page might have higher PageRank themselves. So, even if the number of inlinks might be lower, the quality of inlinks is higher, leading to a higher PageRank for the page. Also, it might just link to itself to have a higher inlink score.

PageRank on Merged Data

- Include a screenshot of the first 20 rows.
- Highlight any differences observed when comparing the PageRank scores from the WT2G_Inlinks data to the merged data set.

```
1. https://www.unwomen.org/en: inlinks - 3817, outlinks - 0, score: 449900423.24099517
2. https://www.unwomen.org/en/information-security: inlinks - 2888, outlinks - 0, score: 449335102.01578355
3. https://donate.unwomen.org/: inlinks - 2584, outlinks - 0, score: 449251562.41910815
4. https://www.unwomen.org/en/terms-of-use: inlinks - 1792, outlinks - 0, score: 446142462.0895435
5. https://www.unwomen.org/en/what-we-do/ending-violence-against-women: inlinks - 1707, outlinks - 234, score: 446097161.41190714
6. https://www.unwomen.org/en/what-we-do/2030-agenda-for-sustainable-development: inlinks - 1702, outlinks - 0, score: 446092612.38835365
7. https://www.unwomen.org/en/digital-library/annual-report: inlinks - 1694, outlinks - 0, score: 446066666.17094487
8. https://www.unwomen.org/en/what-we-do/peace-and-security: inlinks - 1693, outlinks - 3, score: 446065831.2314625
9. https://www.unwomen.org/en/news/editorial-series: inlinks - 1692, outlinks - 0, score: 446064818.17149085
10. https://www.unwomen.org/en/how-we-work/intergovernmental-support: inlinks - 1698, outlinks - 0, score: 446063670.17715496
11. https://www.unwomen.org/en/how-we-work/commission-on-the-status-of-women: inlinks - 1694, outlinks - 5, score: 446063439.8732387
12. https://www.unwomen.org/en/how-we-work/capacity-development-and-training: inlinks - 1689, outlinks - 226, score: 446062755.74515134
13. https://www.unwomen.org/en/about-us/about-un-women: inlinks - 1690, outlinks - 2, score: 446062755.7451511
14. https://www.unwomen.org/en/digital-library/genderterm: inlinks - 1689, outlinks - 0, score: 446062755.7451511
15. https://www.unwomen.org/en/how-we-work/gender-parity-in-the-united-nations: inlinks - 1689, outlinks - 3, score: 446062755.7451511
16. https://www.unwomen.org/en/what-we-do/economic-empowerment: inlinks - 1691, outlinks - 0, score: 446062755.7451511
17. https://www.unwomen.org/en/what-we-do/women-and-girls-with-disabilities: inlinks - 1689, outlinks - 4, score: 446062755.7451511
18. https://www.unwomen.org/en/about-us: inlinks - 1689, outlinks - 0, score: 446062755.7451511
19. https://www.unwomen.org/en/about-us/accountability: inlinks - 1689, outlinks - 0, score: 446062755.7451511
20. https://www.unwomen.org/en/about-us/contact-us: inlinks - 1689, outlinks - 0, score: 446062755.7451511
```

Compared to the WT2G_inlinks dataset, the merged dataset tends to follow the trend more where higher inlinks lead to a higher PageRank score. Also, due to the number of intertwined pages, the score is greatly larger than the WT2G dataset.

HITS Analysis

Hub Scores

- Include a screenshot of the first 20 rows.
- Provide insights into the relationship between Hub scores and the structure of the web graph.

```
https://en.wikipedia.org/wiki/Toleration    0.09462521141161451
https://en.wikipedia.org/wiki/Satya 0.08706933517157825
https://en.wikipedia.org/wiki/Religious_terrorism   0.08645832440737869
https://en.wikipedia.org/wiki/Roman_Catholic    0.07668850598794098
https://en.wikipedia.org/wiki/Mahayana  0.07655319452871812
https://en.wikipedia.org/wiki/Theravada 0.07622866068030451
https://en.wikipedia.org/wiki/Mysticism 0.07309346061704529
https://en.wikipedia.org/wiki/Protestant    0.07270861076524035
https://en.wikipedia.org/wiki/Protestantism 0.07261115440058612
https://en.wikipedia.org/wiki/Hindu 0.07238129063516428
https://en.wikipedia.org/wiki/Atheist   0.07235974208972483
https://en.wikipedia.org/wiki/Ethics_in_religion    0.06881678689162185
https://en.wikipedia.org/wiki/History_of_religion   0.06773926898241313
https://en.wikipedia.org/wiki/Multiculturalism  0.0655456656475711
https://en.wikipedia.org/wiki/God   0.06546114765038971
https://en.wikipedia.org/wiki/Religion_in_ancient_Rome  0.06377190010503099
https://en.wikipedia.org/wiki/Persecution_of_Buddhists  0.06369255686557837
https://en.wikipedia.org/wiki/Judaism   0.06230856706799671
https://en.wikipedia.org/wiki/Religious_conversion  0.06122510535418974
https://en.wikipedia.org/wiki/Thelema   0.06069892910479128
```

Hub score was initially made from the query term, so many of the articles lead to Wikipedia sources. So, we can see the structure of the web graph from the Hub scores since these topics are related (even if the topic of *religion* is not really related to the original search term of the *social justice movement*). The pages above act as a hub of pages with higher authority because each of these Wikipedia pages contains a high number of other Wikipedia articles which will also have higher authority scores.

Authority Scores

- Include a screenshot of the first 20 rows.
- Discuss how Authority scores compare with PageRank and Hub scores, and what this implies about the web pages' importance or relevance.

```
https://en.wikipedia.org/wiki/Wikipedia:General_disclaimer  0.25777251568395837
https://en.wikipedia.org/wiki/Wikipedia:Text_of_the_Creative_Commons_Attribution-ShareAlike_4.0_International_License
https://en.wikipedia.org/wiki/Help:Contents 0.23914542744658654
https://en.wikipedia.org/wiki/Wikipedia:File_upload_wizard  0.236736809708227
https://en.wikipedia.org/wiki/Doi_(identifier)  0.17075933224206313
https://en.wikipedia.org/wiki/Help:Introduction 0.1310934837887919
https://developer.wikimedia.org/    0.1309403592899654
https://www.mediawiki.org/  0.1309403592899654
https://en.wikipedia.org/wiki/Category:Articles_with_short_description  0.12976085622891184
https://en.wikipedia.org/wiki/Wikipedia:Citation_needed 0.12056277151349086
https://en.wikipedia.org/wiki/Help:Authority_control    0.11954292349273697
https://en.wikipedia.org/wiki/Category:Short_description_is_different_from_Wikidata 0.08221338320413979
https://en.wikipedia.org/wiki/Wikipedia:Verifiability   0.07067596271871829
https://en.wikipedia.org/wiki/Category:All_articles_with_unsourced_statements   0.07008639795574854
https://en.wikipedia.org/wiki/Religious_war 0.05823681265660741
https://en.wikipedia.org/wiki/Religious_persecution 0.057685130751891904
https://en.wikipedia.org/wiki/Toleration    0.05748713552262383
https://en.wikipedia.org/wiki/Category:Articles_with_NKC_identifiers    0.05727747524693603
https://en.wikipedia.org/wiki/Category:Articles_with_GND_identifiers    0.05671223681629713
https://en.wikipedia.org/wiki/Religious_terrorism   0.0547674150721436
```

Compared to PageRank, the authority scores lead to many web pages that are more relevant to a certain topic instead of the whole documents crawled. Instead, the above links are linked from many higher-ranked hubs. We can also see that some links are similar to the hub scores because a lot of Wikipedia articles are linked to and from each other, bringing the scores up.

Case Study: PageRank vs. Inlink Count

Select a few pages that have a higher PageRank but a smaller inlink count. For each selected page:

- Identify and describe the other pages that point to them.
- Explain why these pages have a higher PageRank despite a smaller inlink count, focusing on the quality or authority of the incoming links.

By looking at the links below, we can see that there are links pointing to itself, which might drive up the PageRank score. Also, we can speculate that the combined PageRank of the inlinks might be higher, leading to a higher score for these specific pages despite the lower number of inlinks.

WT25-B39-116

- # of Inlinks: 169
- # of Outlinks: 1
- Inlinks:
    - WT25-B38-117 WT25-B40-5 WT25-B40-68 WT25-B40-164 WT25-B40-69 WT25-B40-165 WT25-B40-70 WT25-B40-166 WT25-B40-71 WT25-B40-167 WT25-B40-72 WT25-B40-168 WT25-B40-73 WT25-B40-169 WT25-B40-74 WT25-B40-170 WT25-B40-75 WT25-B40-171 WT25-B40-76 WT25-B40-172 WT25-B40-77 WT25-B40-173 WT25-B40-78 WT25-B40-174 WT25-B40-79 WT25-B40-175 WT25-B40-81 WT25-B40-177 WT25-B40-82 WT25-B40-178 WT25-B40-84 WT25-B40-180 WT25-B40-85 WT25-B40-181 WT25-B40-86 WT25-B40-182 WT25-B40-87 WT25-B40-183 WT25-B40-88 WT25-B40-184 WT25-B40-89 WT25-B40-185 WT25-B40-90 WT25-B40-186 WT25-B40-91 WT25-B40-187 WT25-B40-92 WT25-B40-188 WT25-B40-93 WT25-B40-189

WT25-B40-94 WT25-B40-190 WT25-B40-95 WT25-B40-191 WT25-B40-96
WT25-B40-192 WT25-B40-97 WT25-B40-193 WT25-B40-98 WT25-B40-194
WT25-B40-100 WT25-B40-196 WT25-B40-101 WT25-B40-197
WT25-B40-102 WT25-B40-198 WT25-B40-103 WT25-B40-199
WT25-B40-104 WT25-B40-200 WT25-B40-105 WT25-B40-201
WT25-B40-106 WT25-B40-202 WT25-B40-107 WT25-B40-203
WT25-B40-108 WT25-B40-204 WT25-B39-114 WT25-B39-115
WT25-B39-116 WT25-B39-117 WT25-B40-110 WT25-B40-206
WT25-B40-111 WT25-B40-207 WT25-B40-112 WT25-B40-208
WT25-B40-113 WT25-B40-210 WT25-B40-114 WT25-B40-211
WT25-B40-115 WT25-B40-212 WT25-B40-116 WT25-B40-213
WT25-B40-117 WT25-B40-214 WT25-B40-118 WT25-B40-215
WT25-B40-119 WT25-B40-216 WT25-B40-120 WT25-B40-217
WT25-B40-121 WT25-B40-218 WT25-B40-122 WT25-B40-219
WT25-B40-123 WT25-B40-220 WT25-B40-124 WT25-B40-221
WT25-B40-125 WT25-B40-222 WT25-B40-126 WT25-B40-223
WT25-B40-127 WT25-B40-224 WT25-B40-128 WT25-B40-225
WT25-B40-250 WT25-B40-255 WT25-B40-129 WT25-B40-226
WT25-B40-130 WT25-B40-227 WT25-B40-131 WT25-B40-228
WT25-B40-132 WT25-B40-229 WT25-B40-134 WT25-B40-231
WT25-B40-135 WT25-B40-232 WT25-B40-136 WT25-B40-233
WT25-B40-137 WT25-B40-235 WT25-B40-138 WT25-B40-236
WT25-B40-139 WT25-B40-237 WT25-B40-140 WT25-B40-238
WT25-B40-141 WT25-B40-239 WT25-B40-142 WT25-B40-240
WT25-B40-143 WT25-B40-241 WT25-B40-144 WT25-B40-243
WT25-B40-145 WT25-B40-244 WT25-B39-307 WT25-B39-308 WT25-B40-1
WT25-B40-2 WT25-B40-3 WT25-B40-4 WT25-B39-118 WT25-B39-120
WT25-B39-121 WT25-B39-119 WT25-B39-305 WT25-B39-306
WT25-B39-304 WT25-B39-122 WT25-B39-123

WT23-B21-53

- # of Inlinks: 198
- # of Outlinks: 1
- Inlinks:
    - WT23-B21-43 WT23-B21-233 WT23-B21-210 WT23-B21-222 WT23-B21-223
      WT23-B21-236 WT23-B21-209 WT23-B21-219 WT23-B21-227
      WT23-B21-242 WT23-B21-237 WT23-B21-239 WT23-B21-203
      WT23-B21-241 WT23-B21-240 WT23-B21-238 WT23-B21-217
      WT23-B21-231 WT23-B21-211 WT23-B21-49 WT23-B21-226 WT23-B21-44
      WT23-B21-51 WT23-B21-201 WT23-B21-220 WT23-B21-214 WT23-B21-215
      WT23-B21-202 WT23-B21-221 WT23-B21-206 WT23-B21-225
      WT23-B21-205 WT23-B21-224 WT23-B21-204 WT23-B21-218 WT23-B21-48
      WT23-B21-46 WT23-B21-47 WT23-B21-45 WT23-B21-228 WT23-B21-54
      WT23-B21-50 WT23-B21-234 WT23-B21-55 WT23-B21-232 WT23-B21-56
      WT23-B21-62 WT23-B21-65 WT23-B21-69 WT23-B21-66 WT23-B21-68
      WT23-B21-64 WT23-B21-57 WT23-B21-67 WT23-B21-74 WT23-B21-61
      WT23-B21-58 WT23-B21-63 WT23-B21-60 WT23-B21-70 WT23-B21-71
      WT23-B21-72 WT23-B21-73 WT23-B21-59 WT23-B21-77 WT23-B21-75
      WT23-B21-76 WT23-B21-78 WT23-B21-79 WT23-B21-80 WT23-B21-109

WT23-B21-98 WT23-B21-106 WT23-B21-101 WT23-B21-105 WT23-B21-96
WT23-B21-107 WT23-B21-97 WT23-B21-110 WT23-B21-102 WT23-B21-104
WT23-B21-108 WT23-B21-100 WT23-B21-103 WT23-B21-111 WT23-B21-99
WT23-B21-90 WT23-B21-93 WT23-B21-88 WT23-B21-86 WT23-B21-83
WT23-B21-85 WT23-B21-94 WT23-B21-95 WT23-B21-91 WT23-B21-87
WT23-B21-92 WT23-B21-84 WT23-B21-89 WT23-B21-82 WT23-B21-81
WT23-B21-127 WT23-B21-118 WT23-B21-129 WT23-B21-125
WT23-B21-128 WT23-B21-130 WT23-B21-122 WT23-B21-120
WT23-B21-121 WT23-B21-124 WT23-B21-114 WT23-B21-113
WT23-B21-115 WT23-B21-112 WT23-B21-116 WT23-B21-123
WT23-B21-126 WT23-B21-117 WT23-B21-119 WT23-B21-158
WT23-B21-133 WT23-B21-155 WT23-B21-134 WT23-B21-159
WT23-B21-150 WT23-B21-161 WT23-B21-135 WT23-B21-160
WT23-B21-157 WT23-B21-154 WT23-B21-139 WT23-B21-138
WT23-B21-153 WT23-B21-156 WT23-B21-149 WT23-B21-136
WT23-B21-141 WT23-B21-142 WT23-B21-143 WT23-B21-144
WT23-B21-145 WT23-B21-146 WT23-B21-147 WT23-B21-162
WT23-B21-163 WT23-B21-140 WT23-B21-151 WT23-B21-148
WT23-B21-152 WT23-B21-137 WT23-B21-164 WT23-B21-165
WT23-B21-166 WT23-B21-176 WT23-B21-178 WT23-B21-179
WT23-B21-180 WT23-B21-184 WT23-B21-170 WT23-B21-167
WT23-B21-172 WT23-B21-185 WT23-B21-168 WT23-B21-171
WT23-B21-173 WT23-B21-181 WT23-B21-174 WT23-B21-175
WT23-B21-183 WT23-B21-177 WT23-B21-182 WT23-B21-169
WT23-B21-187 WT23-B21-188 WT23-B21-189 WT23-B21-186
WT23-B21-190 WT23-B21-191 WT23-B21-235 WT23-B21-192
WT23-B21-197 WT23-B21-198 WT23-B21-195 WT23-B21-196
WT23-B21-193 WT23-B21-194 WT23-B21-200 WT23-B21-199
WT23-B21-243 WT23-B21-207 WT23-B21-208 WT23-B21-216
WT23-B21-230 WT23-B21-213 WT23-B21-53 WT23-B21-229 WT23-B21-212

WT24-B26-10

- # of Inlinks: 291
- # of Outlinks: 1
- Inlinks:
    - WT24-B26-10 WT07-B39-205 WT07-B39-162 WT07-B39-81 WT07-B39-177
      WT07-B39-158 WT07-B39-176 WT07-B39-153 WT07-B39-157
      WT07-B39-134 WT07-B39-200 WT07-B39-154 WT07-B39-179
      WT07-B39-160 WT07-B39-170 WT07-B39-203 WT07-B39-178
      WT07-B39-166 WT07-B39-147 WT07-B39-168 WT07-B39-187
      WT07-B39-133 WT07-B39-204 WT07-B39-165 WT07-B39-172 WT07-B39-78
      WT07-B39-82 WT07-B39-131 WT07-B39-186 WT07-B39-202 WT07-B39-175
      WT07-B39-194 WT07-B39-149 WT07-B39-169 WT07-B39-136
      WT07-B39-174 WT07-B39-191 WT07-B39-189 WT07-B39-180
      WT07-B39-198 WT07-B39-152 WT07-B39-171 WT07-B39-137
      WT07-B39-132 WT07-B39-195 WT07-B39-188 WT07-B39-155
      WT07-B39-182 WT07-B39-80 WT07-B39-201 WT07-B39-192 WT07-B39-185
      WT07-B39-74 WT07-B39-115 WT07-B39-124 WT07-B39-116 WT07-B39-117
      WT07-B39-161 WT07-B39-129 WT07-B39-120 WT07-B39-148

WT07-B39-121 WT07-B39-122 WT07-B39-119 WT07-B39-173 WT07-B39-79
WT07-B39-128 WT07-B39-125 WT07-B39-150 WT07-B39-127
WT07-B39-190 WT07-B39-181 WT07-B39-199 WT07-B39-118
WT07-B39-126 WT07-B39-196 WT07-B39-156 WT07-B39-183
WT07-B39-123 WT07-B39-193 WT07-B39-130 WT07-B39-86 WT07-B39-93
WT07-B39-100 WT07-B39-87 WT07-B39-88 WT07-B39-89 WT07-B39-90
WT07-B39-107 WT07-B39-108 WT07-B39-109 WT07-B39-111
WT07-B39-110 WT07-B39-94 WT07-B39-96 WT07-B39-95 WT07-B39-97
WT07-B39-112 WT07-B39-92 WT07-B39-98 WT07-B39-105 WT07-B39-113
WT07-B39-102 WT07-B39-104 WT07-B39-101 WT07-B39-91 WT07-B39-106
WT07-B39-103 WT07-B39-99 WT07-B39-114 WT07-B39-224 WT07-B39-138
WT07-B39-140 WT07-B39-142 WT07-B39-139 WT07-B39-206
WT07-B39-135 WT07-B39-77 WT07-B39-184 WT07-B39-83 WT07-B39-197
WT24-B26-1 WT24-B26-1 WT24-B26-6 WT24-B27-8 WT24-B26-341
WT24-B26-300 WT24-B26-307 WT24-B26-314 WT24-B26-301
WT24-B26-302 WT24-B26-303 WT24-B26-334 WT24-B26-338
WT24-B26-304 WT24-B26-321 WT24-B26-322 WT24-B26-332
WT24-B26-323 WT24-B26-325 WT24-B26-324 WT24-B26-308
WT24-B26-345 WT24-B26-310 WT24-B26-309 WT24-B26-311
WT24-B26-326 WT24-B26-306 WT24-B26-178 WT24-B26-329
WT24-B26-331 WT24-B26-330 WT24-B26-312 WT24-B26-319
WT24-B26-340 WT24-B26-327 WT24-B26-316 WT24-B26-336
WT24-B26-318 WT24-B26-337 WT24-B26-315 WT24-B26-349
WT24-B26-339 WT24-B26-335 WT24-B26-342 WT24-B26-343
WT24-B26-305 WT24-B26-320 WT24-B26-317 WT24-B26-344
WT24-B26-333 WT24-B26-347 WT24-B26-346 WT24-B26-348
WT24-B26-313 WT24-B26-328 WT24-B26-350 WT24-B26-378
WT24-B26-163 WT24-B26-165 WT24-B26-297 WT24-B26-166
WT24-B26-154 WT24-B26-168 WT24-B26-153 WT24-B26-150
WT24-B26-151 WT24-B26-162 WT24-B26-157 WT24-B26-167 WT24-B26-27
WT24-B26-174 WT24-B26-164 WT24-B26-177 WT24-B26-155
WT24-B26-170 WT24-B26-156 WT24-B26-169 WT24-B26-171
WT24-B26-176 WT24-B26-158 WT24-B26-160 WT24-B26-175
WT24-B26-159 WT24-B26-152 WT24-B26-161 WT24-B26-172
WT24-B26-173 WT24-B26-23 WT24-B26-24 WT24-B26-35 WT24-B26-26
WT24-B26-141 WT24-B26-100 WT24-B26-107 WT24-B26-114
WT24-B26-101 WT24-B26-102 WT24-B26-103 WT24-B26-134
WT24-B26-138 WT24-B26-104 WT24-B26-121 WT24-B26-122
WT24-B26-132 WT24-B26-123 WT24-B26-125 WT24-B26-124
WT24-B26-108 WT24-B26-145 WT24-B26-110 WT24-B26-109
WT24-B26-111 WT24-B26-126 WT24-B26-106 WT24-B26-97 WT24-B26-129
WT24-B26-131 WT24-B26-130 WT24-B26-112 WT24-B26-119
WT24-B26-140 WT24-B26-127 WT24-B26-116 WT24-B26-136
WT24-B26-118 WT24-B26-137 WT24-B26-115 WT24-B26-149
WT24-B26-139 WT24-B26-135 WT24-B26-142 WT24-B26-143
WT24-B26-105 WT24-B26-120 WT24-B26-117 WT24-B26-144
WT24-B26-133 WT24-B26-147 WT24-B26-146 WT24-B26-148
WT24-B26-113 WT24-B26-128 WT24-B26-99 WT24-B26-201 WT24-B26-38

WT24-B26-98 WT24-B26-358 WT24-B26-206 WT24-B26-205 WT24-B26-37
WT24-B26-357 WT24-B26-356 WT24-B26-356 WT24-B26-355
WT24-B26-354 WT24-B26-353 WT24-B26-352 WT24-B26-351 WT24-B26-30
WT24-B26-207 WT24-B26-204 WT24-B26-96 WT24-B26-32 WT24-B26-208
WT24-B26-31 WT24-B26-33 WT24-B26-95 WT24-B26-36 WT24-B26-34
WT24-B26-29 WT24-B26-25 WT24-B26-10 WT24-B26-10 WT24-B26-10
WT24-B26-10 WT24-B26-51 WT24-B26-51 WT24-B26-51 WT24-B26-51